Hunter Ellis elliswhunter@gmail.com hjkellis.github.io github.com/hjkellis (703)-953-6963

Experience

Avionics Engineer – Georgia Tech Research Institute
Aug 2025 to Present – Atlanta, GA
- Developing C++ software for collaborative autonomous UAV swarms in a component-based software engineering environment.
- Writing track-to-track fusion algorithms for distributed multi-sensor tracking.
- Built a containerized motion planning service using sampling and graph-based methods for non-holonomic vehicles.
- Benchmarked motion planning algorithms – leveraged The Open Motion Planning Library (OMPL).
- Designed and implemented a track-to-track association system for decentralized tracking systems.
- Refactored legacy software to use Distributed Data Service (DDS) middleware.

Robotics Researcher / M.S. Thesis – Virginia Tech
Aug 2023 to May 2025 – Blacksburg, VA
- Built a 6-DOF robotic arm and an accompanying ROS2–Gazebo simulation environment for training, evaluating, and deploying custom control algorithms.
- Integrated object detection (YOLOv8), natural language processing, symbolic reasoning, and a DDPG-based reinforcement learning policy for robotic manipulation tasks.
- Aided professors in teaching fundamental concepts in linear systems theory and digital signal processing, including Laplace Transforms, Z-Transforms, system stability, and FIR & IIR filter design.
- Assisted with hands-on projects to illustrate and integrate analog and digital filter design and application on breadboards and TI-MSP432 development boards.

Thrust Vector Control Intern – Jacobs Space Exploration Group
May 2024 to Aug 2024 – Huntsville, AL
- Developed hardware and software to test the thrust vector control system of NASA's Mars Ascent Vehicle at the Marshall Space Flight Center.
- Identified and modeled electro-mechanical actuator dynamics for NASA's Active Inertial Load Simulator using Python, MATLAB, and LabVIEW – designed tests to characterize actuator response.
- Integrated IIR filters to reduce high frequency noise from a load cell and LVDT.

Control Research Intern – Grenoble Electrical Engineering Lab
Jun 2023 to Aug 2023 – Grenoble, FR
- Implemented inverter control methods for 4-leg (microgrid) topologies and tested them in Simulink and HITL.
- Simulated neutral point capacitive and balancing topologies using 4-leg inverters in Simulink. Tested PI control, PR control, Clarke and Park Transforms with HITL simulations.

Engineering Intern (NREIP) – Naval Surface Warfare Center
Jun 2022 to Aug 2022 – Bethesda, MD
- Developed concept hospital sea-train designs, estimated fuel consumption, and calculated electrical power loads.

Education

M.S. Computer Engineering
Virginia Tech – 2025
- Focus: Signals and Systems
- Research Group: COOL Autonomy Lab @ UT Austin

B.S. Computer Engineering
B.S. Electrical Engineering
Virginia Tech – 2024
- Focus: Controls, Robotics, and Autonomy

Tools/Libs

ROS2, DDS, GNU/Linux, Git, Qt, Simulink, Docker, FreeRTOS, SOLIDWORKS, Autodesk Inventor

Projects

Robotic Arm – C++, Python, ROS2
- 3D printed robot arm, built using stepper motors and pulleys.
- Integrated ROS2 Jazzy planning and control with a Gazebo Harmonic simulation.

Micro-Quadcopter – Embedded C, ROS2
- Designed and assembled mini-quadcopters with ESP-32 development boards
- Implemented a flight controller in bare-metal C and a corresponding ROS2 communication node.